Visual Analytics Project Report

Exploring Cluster Structures through Interactive Attribute Weighting

A Visual Analytics Interface for Clustering Evaluation

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1 Introduction

The project focuses on the development of a ***Visual Analytics* interface** designed to support the analysis, monitoring, and evaluation of data clustering. The **main goal** of the application is to provide the user with an interactive tool for exploring user-selected datasets, observing the structure of the data, and comparing different clustering approaches.

A central concept in this context is ***dimensionality reduction***, which aims to represent high-dimensional data in a lower-dimensional space while preserving the most relevant structural information. This process is particularly useful for visualization, because it allows complex datasets to be projected onto two-dimensional plots that can be inspected visually. Among the most widely used dimensionality reduction techniques are **Principal Component Analysis (PCA)** and **Multidimensional Scaling (MDS)**. **PCA** projects the data along directions of maximum variance, while **MDS** focuses on preserving the pairwise distances or dissimilarities between items.

In this report, the dimensionality reduction component is based on **MDS**. In **classical MDS**, given a set of target dissimilarities δ_{ij} between items i and j, the objective is to find low-dimensional points $\mathbf{y}_i$ such that their Euclidean distances approximate those dissimilarities. This can be expressed by **minimizing the stress function**

$$\min_{\mathbf{y}_1,\ldots,\mathbf{y}_n} \sum_{i<j} \left(\|\mathbf{y}_i - \mathbf{y}_j\|_2 - \delta_{ij}\right)^2 . \tag{1}$$

The project extends this idea to the case of **non-proportional Euclidean distances**, where the contribution of each non-categorical attribute is controlled by a **coefficient in the interval** $[0, 1]$. These coefficients allow the user to **tune the importance of each attribute** when computing distances between items. If x_{ik} is the value of attribute k for item i and $w_k \in [0, 1]$ is its weight, the weighted non-proportional distance used by the interface can be written as

$$d_w(i,j) = \sqrt{\sum_{k=1}^{p} w_k \left(x_{ik} - x_{jk}\right)^2}$$

Changing the weights modifies the dissimilarity structure given to MDS and therefore changes how the items are positioned in the resulting projection.

2 Methodology

The methodology of the project is organized as a visual analytics pipeline that connects dataset preparation, feature weighting, dimensionality reduction, clustering, evaluation, and interaction. The goal is not only to produce a final clustering result, but also to let the user explore how different attributes influence distances, projections, and cluster separation.

2.1 Dataset Preprocessing

The system allows the user to select one of the default datasets, namely 'wine.csv' and 'iris.csv', or to upload a custom dataset. After a dataset is selected, the application identifies the available attributes and initially sets the first attribute as the default label or reference attribute. The user can then change this selection and decide which attribute should be used as the label for the clustering analysis, as shown in **Figure 1**. Only **non-categorical** attributes are considered in the distance computation and weighting process. The selected **label attribute**, if numeric one, is excluded from the set of weighted features, so that it can be used as a reference for label-based clustering rather than as an input dimension for distance calculation. This preprocessing step makes the analysis adaptable to different datasets and different interpretation goals.

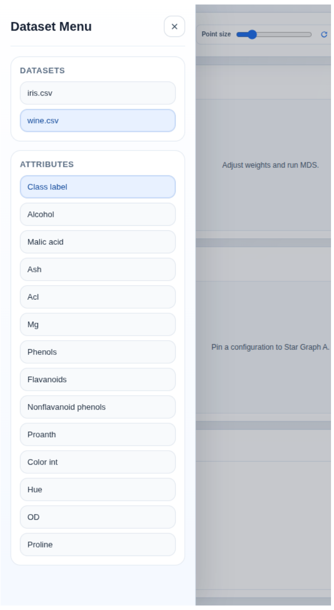

Figure 1: *Attribute-selection interface used to choose the reference label for the analysis.*

2.2 Feature Weighting

The feature weighting model is based on the assumption that different attributes may have different importance in determining the similarity between data items. Each non-categorical attribute is therefore associated with a coefficient $w_k \in [0, 1]$. A value close to zero reduces the influence of the corresponding attribute, while a value close to one increases its contribution to the distance between items.

The user controls these coefficients through the ***Weight Control Panel***. This panel lists the attributes of the selected dataset, excluding the current label or reference attribute. Since the reference attribute can be changed, the set of weighted attributes is updated accordingly. Each coefficient can be adjusted in **steps of 0.05**, allowing a gradual and controlled tuning of the attribute importance. By modifying the weights, the user can analyze how different attribute configurations affect the *MDS projection* and the resulting cluster structure.

2.3 MDS Projection

The dimensionality reduction step is based on Multidimensional Scaling. MDS is used to project the dataset into a two-dimensional space while preserving, as much as possible, the pairwise distances between items. In the main interface, the *Cluster View* is organized around **four scatter plots**, which allow the user to compare different clustering interpretations of the same weighted MDS projection.

Figure 2 shows the initial view of the interface as it appears when the user first opens the application. This view provides the starting point for selecting or uploading a dataset and accessing the available clustering visualizations.

In the **first row**, the two scatter plots show the current MDS projection of the dataset, computed using the attribute weights defined in the ***Weight Control Panel***. Starting from the left, the first plot displays the projection with clusters computed through *K-Means*, while the second plot displays the same projection using the selected *label-based* grouping. To make this comparison consistent, the clusters obtained with *K-Means* and the label-based groups are associated in a unique way by computing the **Jaccard index** between each pair of groups and selecting the best correspondence. This arrangement allows the user to directly compare automatic clustering with the reference labels of the dataset.

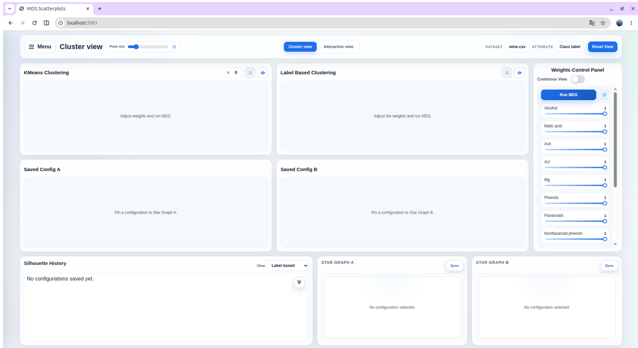

Figure 2: *Initial view of the interface when the application is opened.*

In the **second row**, the interface shows two additional MDS projections associated with saved weighting configurations. These configurations are selected from the *Silhouette History* chart. For each saved configuration, the user can choose whether the corresponding scatter plot should display *K-Means* clustering or *label-based* clustering. In this way, the lower plots make it possible to compare previous weighting configurations and evaluate how different choices of attribute weights affect the projected structure of the data.

2.4 Silhouette Chart

The ***Silhouette Chart*** is used to keep track of the weighting configurations generated during the exploration process. Each saved configuration is represented as a point in the chart, associated with its silhouette value. This allows the user to compare different sets of attribute weights not only visually, but also through a quantitative measure of cluster separation.

The silhouette value summarizes how well the data points fit within their assigned clusters. Higher values indicate more compact and better separated clusters, while lower or negative values suggest ambiguous assignments or overlap between groups. For this reason, the chart acts as a history of the user's experiments and helps identify which weighting configurations improve or worsen the clustering structure.

Figure 3 shows the *Silhouette Chart* used to store and inspect saved weighting configurations during the analysis.

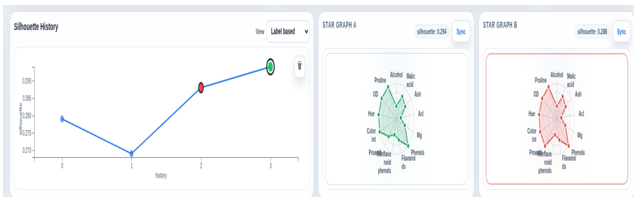

Figure 3: *Silhouette Chart showing saved weighting configurations and their silhouette values.*

When a configuration is saved, the user can retrieve it from the *Silhouette Chart* and compare it with other saved configurations. To display a configuration in a radar graph, the user first selects one of the two available **radar charts** and then clicks on one of the saved configuration points in the *Silhouette Chart*. The selected radar chart is then populated with the attribute coefficients of that configuration.

Since two radar charts are available, the interface supports the parallel visualization of up to two configurations, making it possible to compare two different weight distributions side by side.

Each radar graph is associated with a ***Sync*** button. By pressing *Sync* on one of the two radar graphs, the second row of scatter plots is populated with the MDS points corresponding to the selected saved configuration. In this way, the user can directly connect the silhouette value, the coefficient distribution, and the resulting projected clustering structure.

2.5 Continuous View

The interface also provides a ***Continuous View*** mode for real-time exploration of the weighting model. When this option is active, the user does not need to press the *Run MDS* button after every change. Instead, each modification of an attribute coefficient automatically triggers a live request to the Flask server, which recomputes the weighted distances and returns the updated MDS projection.

The response received from the server is used to update the scatter plots *interactively*, allowing the user to immediately observe how the current weighting configuration changes the spatial distribution of the data. This mechanism makes the exploration more fluid, because the effect of each weight adjustment is visible in real time.

Even in *Continuous View*, configurations are not saved automatically. The user can decide when a specific arrangement of weights is meaningful and store it manually by clicking the dedicated ***Save Configuration*** button. In this way, only relevant configurations are added to the *Silhouette History* chart and later compared through the saved-configuration plots and radar graphs.

2.6 Interactive View

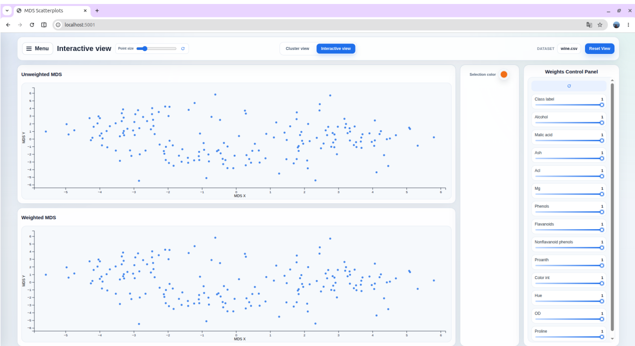

Figure 4: *Initial view of the Interactive View, with the unweighted MDS graph and the weighted MDS graph.*

The *Interactive View* supports direct exploration of the dataset. It contains two main graph windows, both representing MDS projections of the selected dataset. The first graph shows the unweighted MDS projection, while the second shows the weighted one. The weighted projection changes according to the values assigned in the *Weight Control Panel*. These changes are shown **live**: whenever the user modifies a weight, the corresponding plot is updated immediately, allowing the user to observe in real time how the new coefficients affect the position of the points and the overall structure of the projection.

Figure 4 shows the initial layout of the *Interactive View*, with the unweighted and weighted MDS graphs displayed side by side.

A key aspect of this workflow is the possibility of selecting points directly on the MDS plots through a lasso-like interaction tool. The user can draw a selection area around a group of points in one of the two graphs, and the selected points are highlighted in both representations using the chosen color. This linked selection mechanism helps the user compare the same subset of data across the unweighted and weighted MDS views.

Figure 5 shows an example of this interaction. The lasso tool is used to select a subset of points in one MDS graph, and the same points are immediately highlighted in the other graph.

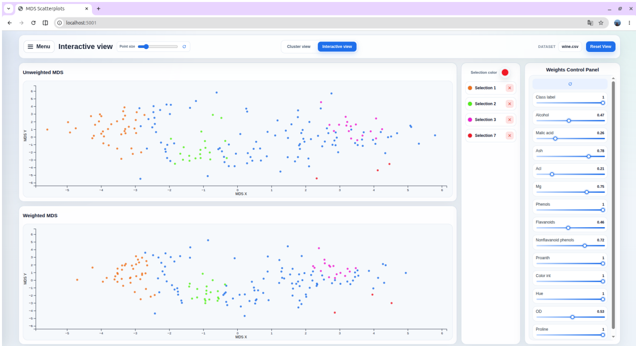

Figure 5: *Example of the lasso tool selecting points in one MDS graph and highlighting the same points in both Interactive View representations.*

The selection can also be deleted and the configuration can be restored by clicking on the refresh button.

2.7 System Architecture

The system architecture is based on two Docker containers, as shown in **Figure 6**. The first container runs a Flask server, which manages the computational part of the application: dataset processing, weighted distance computation, MDS projection, clustering, and silhouette evaluation. The second container runs a Vite server, which provides the front-end resources used by the browser. This separation keeps the data-processing logic on the server side while the browser handles the interactive visual interface.

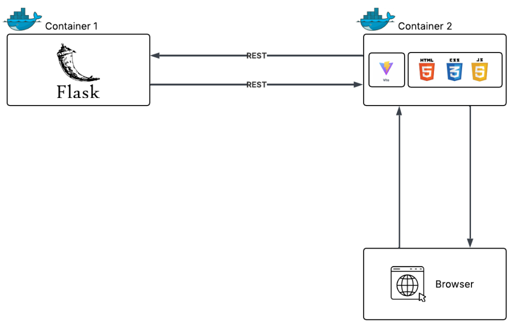

Figure 6: *High-level architecture of the visual analytics application.*

3 Insight

The visualization system helps the user understand which attributes actually drive the similarity between data objects. By changing the weights in the *Weight Control Panel*, the user can observe whether the groups remain stable or whether some points move closer together or farther apart. If a small change in the weight of an attribute produces a clear change in the projection or in the cluster structure, that attribute is likely to be important for the similarity relationships in the dataset.

The interface can also support the identification of clusters, outliers, and hidden relationships. Compact and well-separated groups may suggest meaningful cluster structures, while isolated points or points that frequently change position across configurations may indicate outliers or ambiguous cases. Comparing K-Means clusters with label-based groups further helps the user understand whether the automatic clustering is consistent with the reference labels or whether there are mismatches worth investigating.

However, these insights must be interpreted while considering the limitations of the MDS projection. Since MDS maps high-dimensional data into two dimensions, some distances may be distorted, and points that appear close in the 2D view may not be equally close in the original feature space. The projection can also lose information, especially when the original data contain complex relationships that cannot be represented faithfully in two dimensions.

For this reason, unstable configurations are particularly important to inspect. If small changes in the attribute weights produce very different maps, the resulting visualization may indicate that the dataset structure is sensitive to the selected weighting scheme. In practice, the most reliable interpretation comes from combining multiple signals: point movement, cluster separation, silhouette values, weight distributions, and the stability of the projection across different configurations.

3.1 Use Case 1: Identifying Influential Attributes

A first use case shows how weight tuning can positively influence the clustering result. As shown in **Figure 7**, the silhouette value increases after modifying the attribute coefficients, indicating that the new weighting configuration produces more compact and better separated clusters.

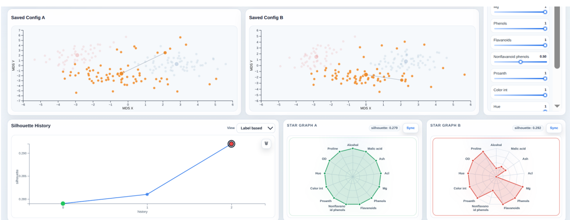

Figure 7: *Example of a weighting configuration that improves the silhouette value and cluster separation.*

This behavior suggests that the selected weights have a direct impact on the quality of the clustering. In particular, attributes assigned lower coefficients tend to spread or disturb the cluster structure less in the weighted distance computation, while more relevant attributes can be emphasized to obtain a clearer separation. The user can therefore compare saved configurations through the *Silhouette Chart* and radar charts to understand which attributes contribute positively to the clustering result.

3.2 Use Case 2: Detecting Outliers and Projection Instability

A second use case concerns the detection of outliers and unstable projection behavior. **Figure 8** shows an example of a possible outlier in the cluster defined by the label-based grouping. The selected point appears problematic because, while the label-based view assigns it to one class, K-Means associates it with a different cluster. This mismatch suggests that the point may be closer, according to the weighted distance structure, to another group than to the one indicated by its reference label.

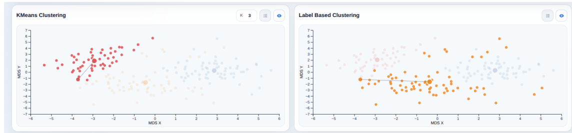

Figure 8: *Example of a possible outlier in the label-based clustering view.*

Figure 9 instead shows an example of projection instability. In this case, the selected point remains in the same cluster in both clustering views. Assuming K-Means as a reliable reference, this suggests that the clustering assignment is stable, but the MDS projection does not place the point meaningfully close to its centroid. Therefore, the issue is not necessarily the clustering result itself, but the way the high-dimensional relationships are represented in the two-dimensional projection.

Figure 9: *Example of projection instability, where the selected point remains in the same cluster but is not projected close to its centroid.*

This use case highlights why the visual interpretation of MDS must be treated carefully. A point can be correctly assigned to a cluster while still appearing visually far from its expected position, because the projection may distort distances or fail to preserve all relationships from the original feature space.